

Implementing External Magnetic Field Maps in Puffin

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1 Introduction

This extension to Puffin allows the magnetic field of an undulator to be supplied as a three-dimensional magnetic field map rather than using one of the built-in analytical undulator models. The feature is intended for simulations using externally generated magnetic fields, for example from magnetostatic calculations or analytical field models. Puffin assumes the supplied magnetic field is valid and performs no checks on its physical consistency.

2 Field Interpolation

The imported magnetic field is defined on a three-dimensional Cartesian mesh, whereas the electron macroparticles move continuously through the simulation domain. Consequently, the magnetic field must be interpolated from the mesh nodes to the position of each macroparticle before the particle equations of motion are evaluated.

Puffin uses trilinear interpolation to determine the magnetic field experienced by each macroparticle. At every integration step, the eight grid points surrounding the particle are identified and their magnetic field values are combined using distance-weighted interpolation. This procedure is performed independently for each magnetic field component (B_x , B_y and B_z).

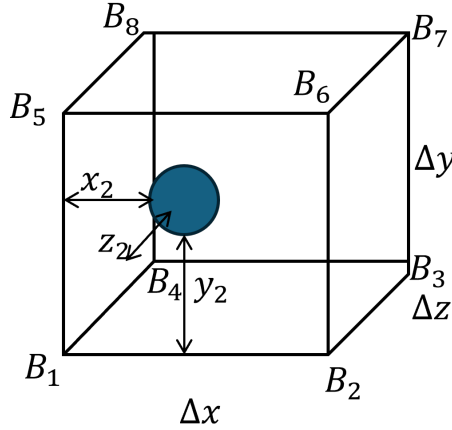


Figure 1: Trilinear interpolation of the magnetic field from the eight surrounding mesh nodes to the electron macroparticle position.

The interpolated magnetic field is given by

$$\mathbf{B}(\mathbf{x}) = \sum_{i=1}^8 w_i \mathbf{B}_i, \quad (1)$$

where $\mathbf{B}_i$ is the magnetic field at the i th surrounding mesh node and w_i are the corresponding trilinear interpolation weights.

The accuracy of the interpolated field depends on the spatial resolution of the supplied field map. Smooth magnetic field distributions can generally be represented accurately using relatively modest mesh resolutions, while fields containing rapid spatial variations may require a finer mesh.

3 Magnetic Field Input

The external magnetic field is supplied as a single HDF5 file containing six datasets. Three datasets contain the magnetic field components (B_x , B_y and B_z), while the remaining three datasets (x , y and z) define the corresponding coordinate axes of the field map. Figure 2 shows the required HDF5 file structure.

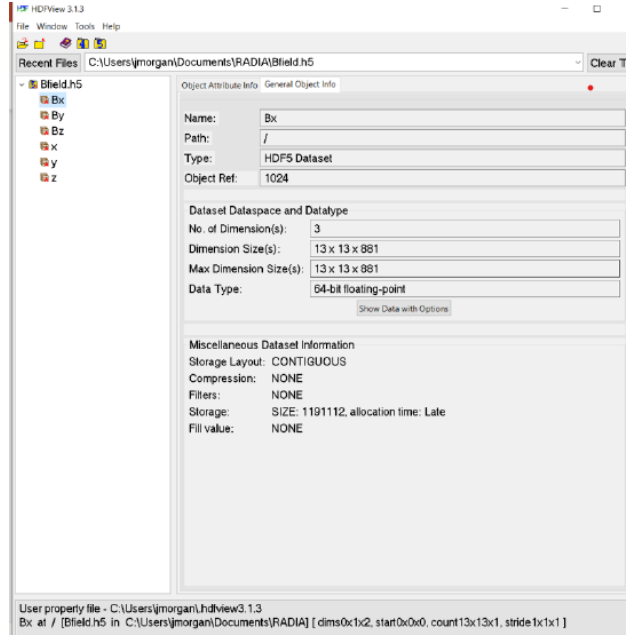


Figure 2: Required HDF5 file structure for importing external magnetic fields. The file must contain the magnetic field components (B_x , B_y , and B_z) together with the corresponding coordinate arrays (x , y , and z) defining the Cartesian mesh.

The magnetic field is assumed to be defined on a uniformly spaced Cartesian mesh. The B_x , B_y and B_z datasets must therefore have dimensions consistent with the lengths of the x , y and z coordinate arrays.

The transverse coordinate arrays (x and y) are assumed to be centred symmetrically about zero, while the longitudinal coordinate array (z) is assumed to begin at zero. These coordinate arrays are used to determine the spatial extent and grid spacing of the imported magnetic field and should therefore accurately describe the mesh on which the field components are defined.

An example Jupyter notebook is provided with the implementation to generate HDF5 magnetic field files from analytical field expressions. This notebook may be readily adapted to generate magnetic field maps from external magnetostatic software, such as RADIA, or from other sources.

4 Lattice Input

The external magnetic field is specified using the UNF lattice element. This element behaves in the same manner as Puffin’s existing analytical undulator elements and may be used alongside quadrupoles, drifts, phase shifters and other lattice elements. Replacing an analytical undulator with a UNF element requires no other modifications to the lattice.

Figure 3 shows the equivalent analytical and imported field definitions for a planar undulator. Replacing the analytical UN element with the UNF element imports the magnetic field from the specified HDF5 file while leaving the remainder of the lattice unchanged.

```

11 ! External magnetic field imported from HDF5
12
13 UNF 'planar_validation.h5' 90 30
14
15
16 ! The parameter for the external magnetic feild identifier UNF
17
18 ! Name of the hdf5 file containing the field
19
20 ! Number of periods in the undulator.
21
22 ! Number of intergration steps per period

```

Figure 3: Example lattice definitions for the analytical planar undulator and the equivalent imported magnetic field.

The UNF lattice element has the form

UNF '<field_file.h5>' Nperiods Nsteps

where

- <field_file.h5> is the HDF5 file containing the imported magnetic field.
- Nperiods specifies the length of the undulator as the number of undulator periods.
- Nsteps specifies the number of integration steps per undulator period used by Puffin.

Unlike the analytical undulator elements, the undulator length is not inferred from the imported magnetic field file. Instead, Puffin propagates the electrons through the imported field over a distance of $N_{\text{periods}} \times \lambda_u$, sampling the magnetic field at each integration step.

Consequently, the imported field map does not need to terminate exactly at the specified undulator length. If the HDF5 file contains a longer magnetic field map than required, only the portion corresponding to the specified lattice length is used. It is the responsibility of the user to ensure that the imported magnetic field is defined over the region required for the simulation.

5 Current Limitations

The external magnetic field implementation assumes that the supplied magnetic field map is valid and compatible with the simulation being performed. No additional validation of the imported field is currently performed.

In particular:

- The imported magnetic field is assumed to satisfy Maxwell's equations. Puffin performs no checks on the physical consistency of the supplied field map.
- Puffin does not verify that the electron beam remains within the spatial extent of the imported magnetic field. It is the responsibility of the user to ensure that the magnetic field map covers the full region explored by the electron beam throughout the simulation.

- Puffin does not verify that the longitudinal sampling of the imported magnetic field is sufficient to accurately model the FEL interaction. Similarly, no checks are performed to ensure that the imported field is compatible with the radiation wavelength and numerical resolution used in the simulation. Users are responsible for selecting an appropriate magnetic field resolution for their application.
- The imported magnetic field is evaluated using trilinear interpolation between the supplied grid points. The accuracy of the simulation therefore depends on the spatial resolution of the supplied magnetic field map.